



BRAC UNIVERSITY

Department of Computer Science & Engineering

Quiz 4

CSE 350: Digital Electronics and Pulse Techniques



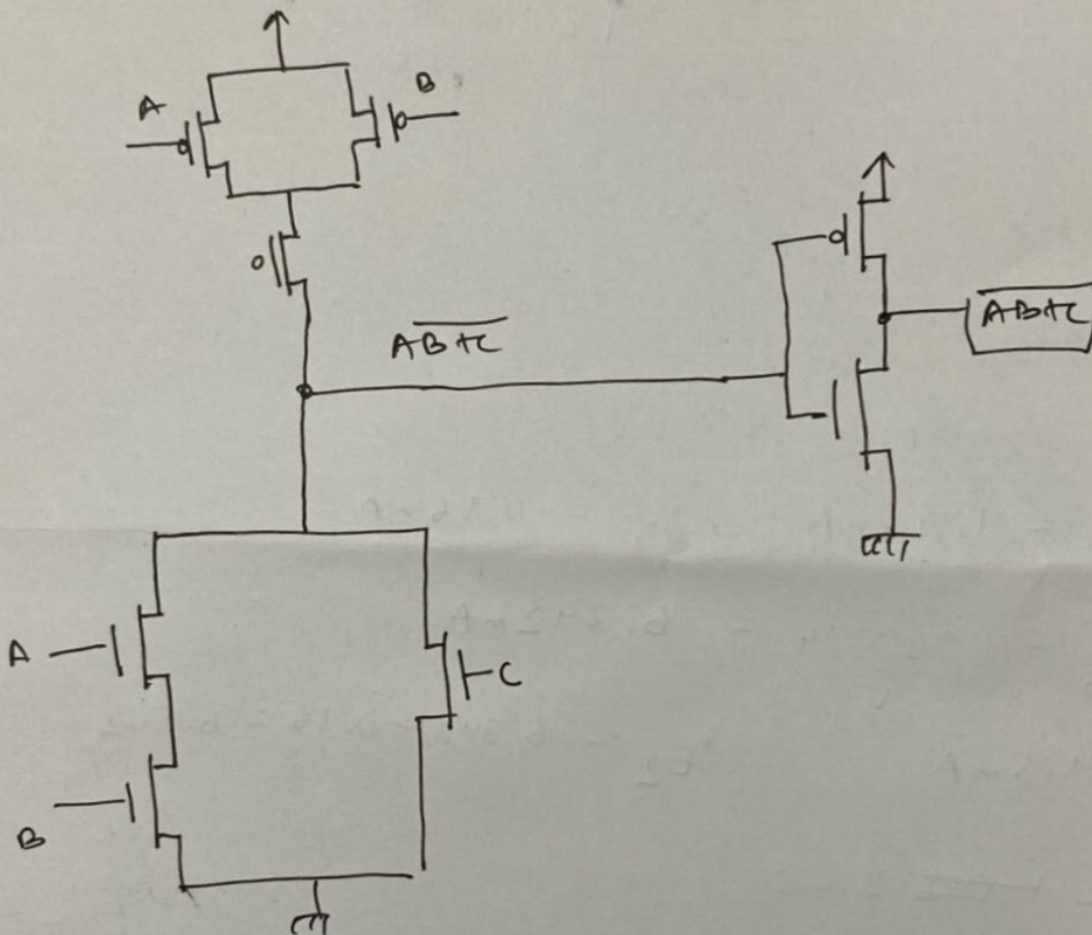
Total Marks: 30

TSE, Spring 26.

Time: 45 Minutes

Name:	ID:	Section:
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1. Draw the logic function $Y = AB + C$ using CMOS logic [10]



2. For the following circuit, given that Logic High = 10V and Logic Low = 0.2 V,

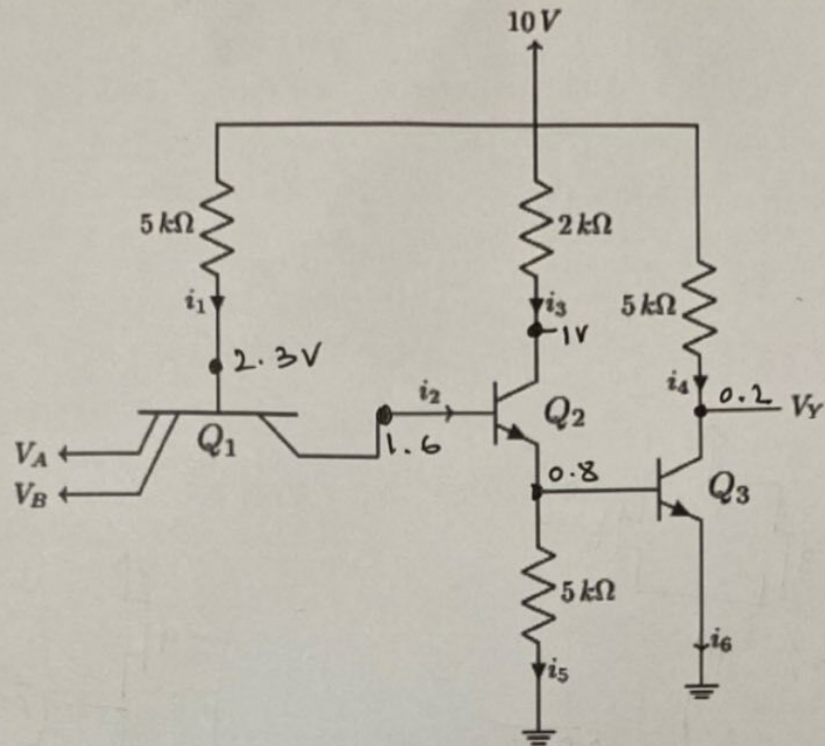


Figure 1: TTL NAND gate

(a) If $i_6 = 8.302\text{mA}$, find the value of reverse beta, β_R of Q_1 . [20]

$$i_4 = 1.96\text{mA} \quad , \quad i_5 = 0.16\text{mA}$$

$$i_{B3} = i_6 - i_4 = 6.342\text{mA}$$

$$i_3 = 4.5\text{mA}$$

$$i_{E2} = 6.342 + 0.16 = 6.502$$

$$\therefore i_2 = \cancel{1.842} \quad 2.002$$

$$i_1 = 1.54$$

$$\therefore i_{QA} = i_{RB} = \frac{2.002 - (\cancel{1.842} - 1.54)}{2} = \frac{0.231}{2} = \cancel{0.302}$$

$$\therefore \beta_R = \frac{\cancel{0.702}}{\cancel{1.54}} = \frac{0.231}{1.54}$$

$$\boxed{\beta_R = 0.15}$$